13.3: Meiosis reduces the number of chromosome sets from diploid to haploid

Several steps of meiosis closely resemble corresponding steps in mitosis. Meiosis, like mitosis, is preceded by interphase, which includes S phase (the duplication of chromosomes). However, this is followed by not one but two consecutive cell divisions, called **meiosis I** and **meiosis II**. These two divisions result in four daughter cells (rather than the two daughter cells of mitosis), each with only half as many chromosomes as the parent cell—one set, rather than two.

The Stages of Meiosis

The overview of meiosis in **Figure 13.7** shows, for a single pair of homologous chromosomes in a diploid cell, that both members of the pair are duplicated and the copies sorted into four haploid daughter cells. Recall that sister chromatids are two copies of *one* chromosome, closely associated all along their lengths; this association is called *sister chromatid cohesion*. Together, the sister chromatids make up one duplicated chromosome (see Figure 13.4). In contrast, the two chromosomes of a homologous pair are individual chromosomes that were inherited from each parent. Homologs appear alike in the microscope, but they may have different versions of genes at corresponding loci; each version is called an *allele* of that gene (see Figure 14.4). For example, one chromosome might have an allele for freckles, but the homologous chromosome may have an allele for the absence of freckles at the same locus. Homologs are not associated with each other in any obvious way except during meiosis.

Figure 13.7

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Overview of meiosis: how meiosis reduces chromosome number.
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Diploid cell with one pair homologous chromosomes goes through Interphase, Meiosis 1 and Meiosis 2. Four haploid daughter cells result, each with one chromosome (two with paternal homologs; 2 maternal).

After the chromosomes duplicate in interphase, the diploid cell divides twice, yielding four haploid daughter cells. This overview tracks just one pair of homologous chromosomes, which for the sake of simplicity are drawn in the condensed state throughout. (They would not normally be condensed during interphase.)

Interphase: Diploid cell drawn with one pair of homologous chromosomes: one maternal (red) and one paternal (blue) homolog. Arrows indicate duplication of chromosomes, with the result being a diploid cell with duplicated chromosomes: maternal and paternal homologs both in sister chromatid form (sister chromatids attached at centromere).

Meiosis 1: Homologous chromosomes separate. The result is two haploid cells, each with a duplicated chromosome (paternal sister chromatids in one cell, maternal sister chromatids in the other cell. Sister chromatids remain attached at the centromere).

Meiosis 2: Sister chromatids separate. The result is four haploid daughter cells: 2 from the paternal chromatids, each cell now containing a single paternal chromosome, and 2 from the maternal chromatids, each cell now containing a single maternal chromosome. The four haploid daughter cells contain unduplicated chromosomes.

DRAW IT *Redraw the cells in this figure using a simple double helix to represent each DNA molecule.*

*For suggested answer, see*Appendix A.*

Animation

Overview of Meiosis

image

image

**

Figure 13.8 describes in detail the stages of the two divisions of meiosis for an animal cell whose diploid number is 6. Study this figure thoroughly before going on.

Figure 13.8 Interactive Visual Activity

Exploring Meiosis in an Animal Cell

Interact with this figure in the Figure 13.8 Interactive Visual Activity: Exploring Meiosis in an Animal Cell.

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Figure 13.8

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Exploring Meiosis in an Animal Cell
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MEIOSIS 2: Separates sister chromatids and is made up of Prophase 2, Metaphase 2, Anaphase 2, Telophase 2 and Cytokinesis.

During another round of cell division, the sister chromatids finally separate in Anaphase 2; four haploid daughter cells result, containing unduplicated chromosomes.

Prophase 2:

- A spindle apparatus forms.
- In late prophase 2 (not shown here), chromosomes, each still composed of two chromatids associated at the centromere, is moved by microtubules toward the metaphase 2 plate.

Metaphase 2:

- The chromosomes are positioned at the metaphase plate as in mitosis.
- Because of crossing over in meiosis 1, the two sister chromatids of each chromosome are not genetically identical.
- The kinetochores of sister chromatids are attached to microtubules extending from opposite poles.

Anaphase 2:

- Breakdown of proteins holding the sister chromatids together at the centromere allows the chromatids to separate. The chromatids move toward opposite poles as individual chromosomes.

Telophase 2 and cytokinesis:

- Nuclei form, the chromosomes begin decondensing, and cytokinesis occurs.
- The meiotic division of one parent cell produces four daughter cells, each with a haploid set of (unduplicated) chromosomes.
- The four daughter cells are genetically distinct from one another and from the parent cell.

MAKE CONNECTIONS *Look at**Figure 12.7**and imagine the two daughter cells undergoing another round of mitosis, yielding four cells. Compare the number of chromosomes in each of those four cells, after mitosis, with the number in each cell in*Figure 13.8*, after meiosis. Explain how the process of meiosis results in this difference, even though meiosis also includes two cell divisions.*

*For suggested answer, see*Appendix A*.*

Video

Meiosis I in Sperm Formation

image

Play

image

Play

BioFlix® Animation

Meiosis

image

Play

image

**

Animation

Meiosis

image

Play

image

Play

Crossing Over and Synapsis During Prophase I

Prophase I of meiosis is a very busy time. The prophase I cell shown in Figure 13.8 is at a point fairly late in prophase I, when pairing of homologous chromosomes, crossing over, and chromosome condensation have already taken place. The sequence of events leading up to that point is shown in more detail in **Figure 13.9**.

Figure 13.9

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Crossing over and synapsis in prophase I: a closer look.
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A diagram of crossing over and synapsis in a series of four steps.

Steps are as follows.

- After interphase, the chromosomes have been duplicated, and sister chromatids are held together by proteins called cohesins (purple). Each pair of homologs is associated along their length. The DNA molecules of two non-sister chromatids are broken at precisely corresponding points. The chromatin of the chromosomes starts to condense.
- A zipper-like protein complex, the synaptonemal complex (green), begins to form, attaching one homolog to the other. The chromatin continues to condense.
- a. The synaptonemal complex is fully formed; the two homologs are said to be in synapsis. During synapsis, the DNA breaks are closed up when each broken end is joined to the corresponding segment of the non-sister chromatid, producing crossovers. After the synaptonemal complex disassembles, the homologs move slightly apart from each other but remain attached because of sister chromatid cohesion, even though some of the DNA may no longer be attached to its original chromosome. The points of attachment where crossovers have occurred show up as chiasmata. The chromosomes continue to condense as they move toward the metaphase plate.

VISUAL SKILLS *Label the four DNA breaks in step 2.*

*For suggested answer, see*Appendix A.**

After interphase, the chromosomes have been duplicated and the sister chromatids are held together by proteins called *cohesins*. **STEP 1** Early in prophase I, the two members of a homologous pair associate loosely along their length. Each gene on one homolog is aligned precisely with the corresponding allele of that gene on the other homolog. The DNA of two nonsister chromatids—one maternal and one paternal—is broken by specific proteins at precisely matching points. **STEP 2** Next, the formation of a zipper-like structure called the **synaptonemal complex** holds one homolog tightly to the other. **STEP 3** During this association, called **synapsis**, the DNA breaks are closed up so that each broken end is joined to the corresponding segment of the *nonsister* chromatid. Thus, a paternal chromatid is joined to a piece of maternal chromatid beyond the crossover point, and vice versa.

STEP 4 These points where crossing over has just occurred become visible as chiasmata (singular, **chiasmata**) after the synaptonemal complex disassembles and the homologs move slightly apart from each other. The homologs remain attached because sister chromatids are still held together by sister chromatid cohesion, even though some of the DNA may no longer be attached to its original chromosomal DNA. At least one crossover per chromosome must occur in order for the homologous pair to stay together as it moves to the metaphase I plate, for reasons that will be explained shortly.

A Comparison of Mitosis and Meiosis

Figure 13.10 summarizes the key differences between meiosis and mitosis in diploid cells. Meiosis produces four cells and reduces the number of chromosome sets from two (diploid) to one (haploid), whereas mitosis produces two cells and conserves the number of chromosome sets. Meiosis produces cells that differ genetically from the parent cell and from each other, whereas mitosis produces daughter cells that are genetically identical to their parent cell and to each other.

Figure 13.10

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A comparison of mitosis and meiosis.
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Comparison of Mitosis and Meiosis during Prophase, Metaphase, Anaphase and Telophase.

Two tables are given.

The first table is as follows.

Mitosis	Meiosis
Prophase: Parent cell (before chromosome duplication) is shown (2 n equals to 6). Chromosome duplication is initiated and duplicated chromosomes are formed (two sister chromatids)	(Meiosis 1) Prophase 1: Parent cell (before chromosome duplication) is shown (2 n equals to 6). Chromosome duplication is initiated and homologous chromosome pairs are formed which are held together by chiasma (site of crossing over) and sister chromatid cohesion.
Metaphase: Individual chromosomes line up at the metaphase plate.	Metaphase 1: Pairs of homologous chromosomes line up at the metaphase plate.
Anaphase, Telophase: Sister chromatids separate during anaphase and 2 daughter cells of mitosis are formed (2 n)	Anaphase 1, Telophase 1: Homologs separate during anaphase 1; sister chromatids remain attached at centromere. Daughter cells of meiosis 1 are formed (Haploid n equals to 3)
Anaphase, Telophase: Sister chromatids separate during anaphase and 2 daughter cells of mitosis are formed (2 n)	Sister chromatids separate during anaphase 2 creating daughter cells (n) of meiosis 2

The second table is as follows.

Summary
Property

DNA replication
Number of divisions
Synapsis of homologous chromosomes
Number of daughter cells and genetic composition
Role in animals, fungi, and plants

DRAW IT *Could any other combinations of chromosomes be generated during meiosis II from the specific cells shown in telophase I? Explain. (Hint: Draw the cells as they would appear in metaphase II.)*

*For suggested answer, see*Appendix A.**

Figure 13.10 Walkthrough

A comparison of mitosis and meiosis

Watch an explanation of this figure in the Figure 13.10 Walkthrough: A comparison of mitosis and meiosis.

image

Play

image

Three events unique to meiosis occur during meiosis I:

- **Synapsis and crossing over.** During prophase I, duplicated homologs pair up and crossing over occurs as described previously and in Figure 13.9. Synapsis and crossing over do not occur during prophase of mitosis.
- **Alignment of homologous pairs at the metaphase plate.** At metaphase I of meiosis, pairs of homologs are positioned at the metaphase plate, rather than individual chromosomes, as in metaphase of mitosis.
- **Separation of homologs.** At anaphase I of meiosis, the duplicated chromosomes of each homologous pair move toward opposite poles, but the sister chromatids of each duplicated chromosome remain attached. In anaphase of mitosis, by contrast, sister chromatids separate.

Sister chromatids stay together due to sister chromatid cohesion, mediated by cohesin proteins. In mitosis, this attachment lasts until the end of metaphase, when enzymes cleave the cohesins, freeing the sister chromatids to move to opposite poles of the cell. In meiosis, sister chromatid cohesion is released in two steps: cohesion of *arms* at the start of anaphase I and cohesion of *centromeres* at anaphase II. In metaphase

I, the two homologs of each pair are held together because there is still cohesion between sister chromatid arms in regions beyond points of crossing over, where stretches of sister chromatids now belong to different chromosomes. The combination of crossing over and sister chromatid cohesion along the arms results in the formation of a chiasma. Chiasmata hold homologs together as the spindle forms for the first meiotic division. At the onset of anaphase I, the release of cohesion along sister chromatid arms allows homologs to separate. At anaphase II, the release of sister chromatid cohesion at the centromeres allows the sister chromatids to separate. Thus, sister chromatid cohesion and crossing over, acting together, play an essential role in the lining up of chromosomes by homologous pairs at metaphase I.

Meiosis I reduces the number of chromosome sets: from two (diploid) to one (haploid). During the second meiotic division, sister chromatids separate, producing four haploid daughter cells. The mechanisms for separating sister chromatids in meiosis II and mitosis are virtually identical. The molecular basis of chromosome behavior during meiosis continues to be a focus of intense research. In the **Scientific Skills Exercise**, you can work with data that track the amount of DNA in cells as they progress through meiosis.

Scientific Skills Exercise

Making a Line Graph and Converting Between Units of Data

AP Science Practices 2, 4, 5

How Does DNA Content Change as Budding Yeast Cells Proceed Through Meiosis? When nutrients are low, cells of the budding yeast (*Saccharomyces cerevisiae*) exit the mitotic cell cycle and enter meiosis. In this exercise, you will track the DNA content of a population of yeast cells as they progress through meiosis.

Budding yeast cells look similar to eggs.

Budding yeast cells

How the Experiment Was Done Researchers grew a culture of yeast cells in a nutrient-rich medium and then transferred the cells to a nutrient-poor medium to induce meiosis. At different times after induction, the DNA content per cell was measured in a sample of the cells, and the average DNA content per cell was recorded in femtograms (fg; 1 femtogram=1×10−15 gram).

Data from the Experiment

A table lists the time after induction of meiosis (hour) and the average amount of D N A per cell (femtograms).

The table is as follows.

Time after induction of meiosis in hours	Average amount of D N A per cell in femtograms
0	24
1	24
2	40
3	47
4	47.5
5	48
6	48
7	47.5
7.5	25
8	24
9	23.5
9.5	14
10	13
11	12.5
12	12
13	12.5
14	12

Further Reading G. Simchen, Commitment to meiosis: What determines the mode of division in budding yeast? *BioEssays* 31:169–177 (2009).

INTERPRET THE DATA

- First, set up your graph.
- Place the labels for the independent and dependent variables on the appropriate axes, followed by units of measurement in parentheses. Explain your choices.

- Add tick marks and values for each axis. Explain your choices. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Because the variable on the x-axis varies continuously, it makes sense to plot the data on a line graph (if you are using Excel, choose “Scatter”).
- Plot each data point from the table onto the graph.
- Connect the data points with line segments.
- Most of the yeast cells in the culture were in G1 of the cell cycle before being moved to the nutrient-poor medium.
- How many femtograms of DNA are there in each yeast cell in G1? Estimate this value from the data in your graph.
- How many femtograms of DNA should be present in each cell in G2? (See Concept 12.2 and Figure 12.6.) At the end of meiosis I (MI)? At the end of meiosis II (MII)? (See Figure 13.7.)
- Using these values as a guideline, distinguish the different phases by inserting vertical dashed lines in the graph between phases and label each phase (G1, S, G2, MI, MII). You can figure out where to put the dividing lines based on what you know about the DNA content of each phase (see Figure 13.7).
- Think carefully about the point where the line at the highest value begins to slope downward. What specific point of meiosis does this “corner” represent? What stage(s) correspond(s) to the downward sloping line?
- Given the fact that 1 fg of DNA=9.78×105 base pairs (on average), you can convert the amount of DNA per cell to the length of DNA in numbers of base pairs.
- Calculate the number of base pairs of DNA in the haploid yeast genome. Express your answer in millions of base pairs (Mb), a standard unit for expressing genome size. Show your work.
- How many base pairs per minute were synthesized during the S phase of these yeast cells?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.